Unit03 : Data Warehousing Architecture

CONTENTS

Objectives

After this lecture, you will be able to

- Understand the concept of operational data.
- Learn the architecture and applications of operational data stores.
- Learn the functions and architecture of various process managers.
- Understand the functions and architecture of the query manager.
- Learn the different types of end-user access tools.
- Understand the concept of the architecture model.
- Know the difference in working of 2-tier,3-tier, and 4-tier architecture.

Introduction

Data warehouse provides architectures and tools for business executives to systematically organize, understand, and use their data to make strategic decisions. In the last several years, many firms have spent millions of dollars in building enterprise-wide data warehouses as it is assumed a way to keep customers by learning more about their needs.

In simple terms, a data warehouse refers to a database that is maintained separately from an organization's operational databases. Data warehouse systems allow for the integration of a variety of application systems. They support information processing by providing a solid platform of consolidated, historical data for analysis.

3.1 Operational Data and Datastore

Operational Data is exactly what it sounds like - Data that is produced by your organization's day-to-day operations. The Operational Database is the source of information for the data warehouse.It includes detailed information used to run the day-to-day operations of the business. Operational Database Management Systems also called as OLTP are used to manage dynamic data in real-time.

Operational data stores (ODS) are data repositories that store a snapshot of an organization's current data. This is a highly volatile data repository that is ideally suited for real-time analysis.

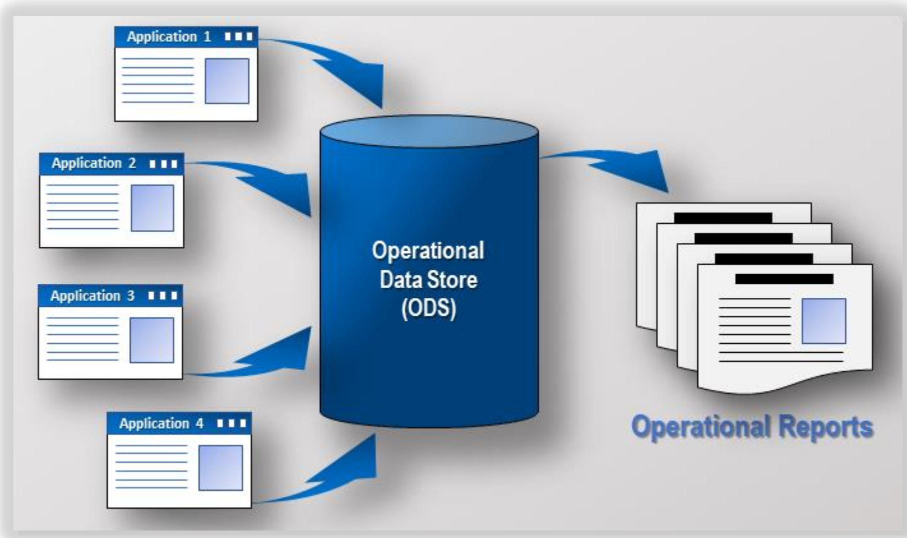

Figure 1: Operational Data Store

An ODS is an integrated database of operational data. Its sources include legacy systems, and it contains current or near-term data. An ODS may contain 30 to 60 days of information, while a data warehouse typically contains years of data. ODSs are used in some data warehouse architectures to provide near-real-time reporting capability if the Data Warehouse's loading time or architecture prevents it from being able to provide near-real-time reporting capability. The ODS then only provides access to the current, fine-grained, and non-aggregated data, which can be queried in an integrated manner without burdening the transactional systems. However, more complex analyses requiring high-volume historical and/or aggregated data are still conducted on the actual data warehouse.

Characteristics of Operational Data Store Systems

The following are the characteristics of Operational Data Stores(ODS):

- ODS systems are highly available and fault-tolerant.
- They occupy less space due to the compression of data and operations.
- ODS systems host configurable, easily accessible, and fast real-time comprehensive data.
- ODS systems are connected to one or more data sources.
- They do not host large amounts of historical data, and thus cannot handle huge data transactions.
- An ODS system makes the creation of back-ups and recovery processes effortlessly since the size of the data is small.

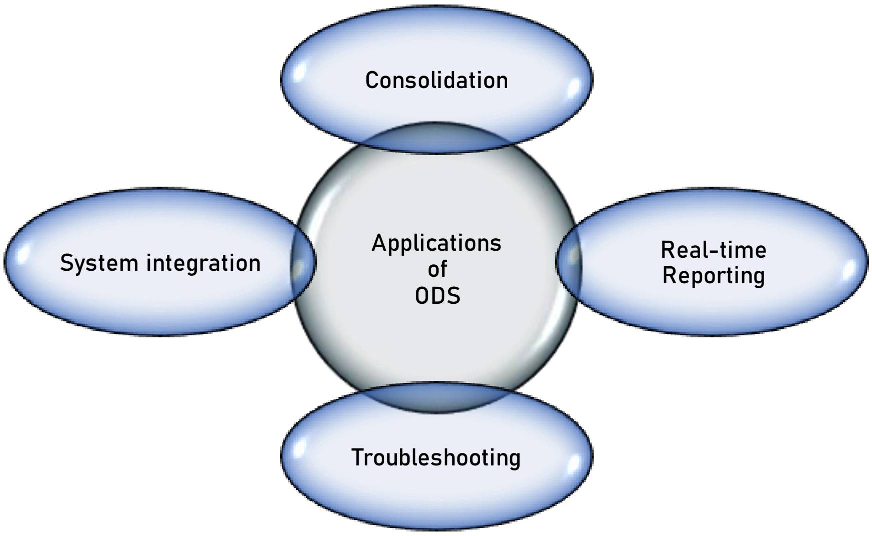

Figure 2: Applications of ODS

- **Consolidation**: The ODS approach can bring together disparate data sources into a single repository. It lacks the benefits of other repositories such as data lakes and data warehouses, but an operational data store has the advantage of being fast and light. ODS can consolidate data from different sources, different systems, or even different locations.
- **Real-time reporting:** An operational data store will generally hold very recent versions of business data. Combined with the right BI tools, businesses can perform real-time BI tasks, such as tracking orders, managing logistics, and monitoring customer activity.
- **Troubleshooting**: The current state view of ODS makes it easier to identify and diagnose issues when they occur. For The ODS will hold both versions of the data, allowing for easy comparison between the two systems. Automated processes can spot these problems and take action.
- **System integration:** Integration requires a continuous flow of data between systems, and ODS can provide the platform for this kind of exchange. It's possible to build business rules on an ODS so that data changes in one system triggers a corresponding action on another system.

A user might create an order on the e-commerce system, which should create a corresponding order on the logistics system. But this might have the wrong details due to an integration error.

Working of ODS

The extraction of data from source databases needs to be efficient, and the quality of records needs to be maintained. Since the data is refreshed generally and frequently, suitable checks are required to ensure the quality of data after each refresh. An ODS is a read-only database other than regular refreshing by the OLTP systems. Customers should not be allowed to update ODS information.

Populating an ODS contains an acquisition phase of extracting, transforming, and loading information from OLTP source systems. This procedure is ETL. Completing populating the database, analyze for anomalies, and testing for performance is essential before an ODS system can go online.

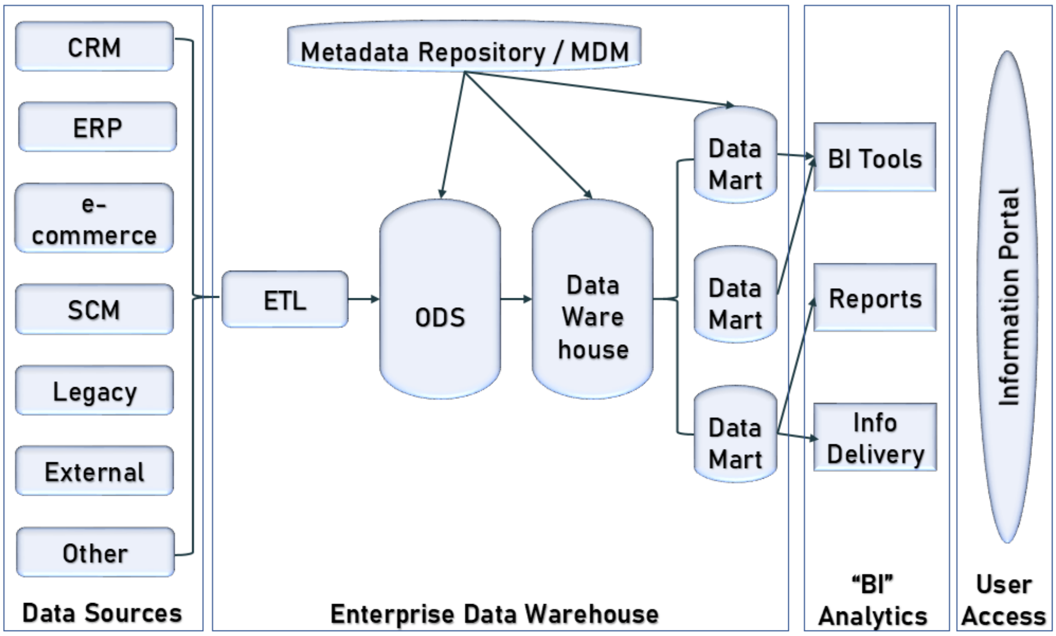

Figure 3: Working of Operational Datastore

Difference between Operational Data Source and Data Warehouse

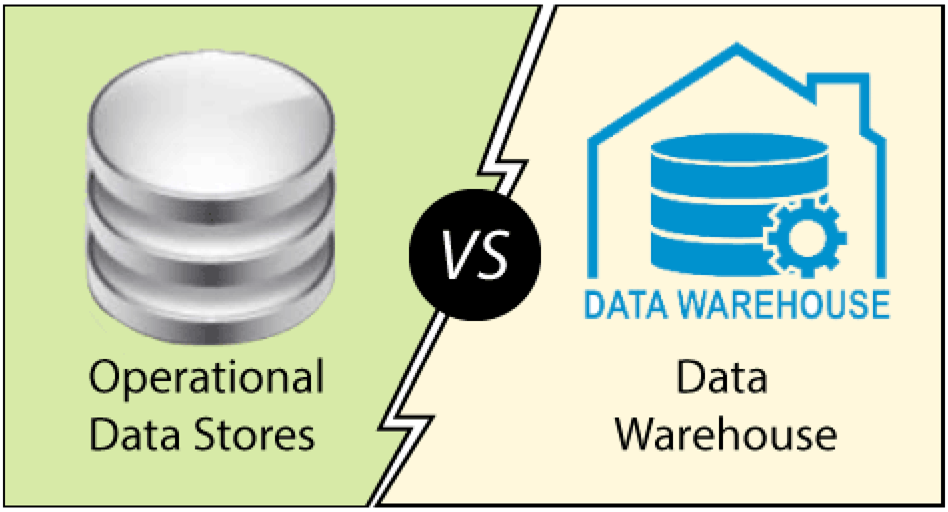

Figure 4: Difference in ODS and Data warehouse

Operational Data Stores	**Data Warehouse**
ODS means **operational** reporting and supports current or near real-time reporting requirements.	A **data warehouse** is intended for historical and trend analysis, usually reporting on a large volume of data.
An **ODS** consist of only a short window of **data**.	A **data warehouse** includes the entire history of **data**.
It is typically detailed data only.	It contains summarized and detailed data.

It is used for detailed decision-making and operational reporting.	It is used for long-term decision-making and management reporting.
It is used at the operational level.	It is used at the managerial level.
It serves as a conduit for data between operational and analytics systems.	It serves as a repository for cleansed and consolidated data sets.
It is updated often as the transactions system generates new data.	It is usually updated in batch processing mode on a set scheme
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You know about store room and warehouse. Exactly what the difference between ODS and data warehouse? Explain with the suitable of suitable example.

3.2 Process Manager

Process managers are responsible for maintaining the flow of data both into and out of the data warehouse. There are three different types of process managers:

a. Load manager

b. Warehouse manager

c. Query manager

Data Warehouse Load Manager

The load manager performs the operations required to extract and load the data into the database. The size and complexity of a load manager vary between specific solutions from one data warehouse to another.

The architecture of load manager

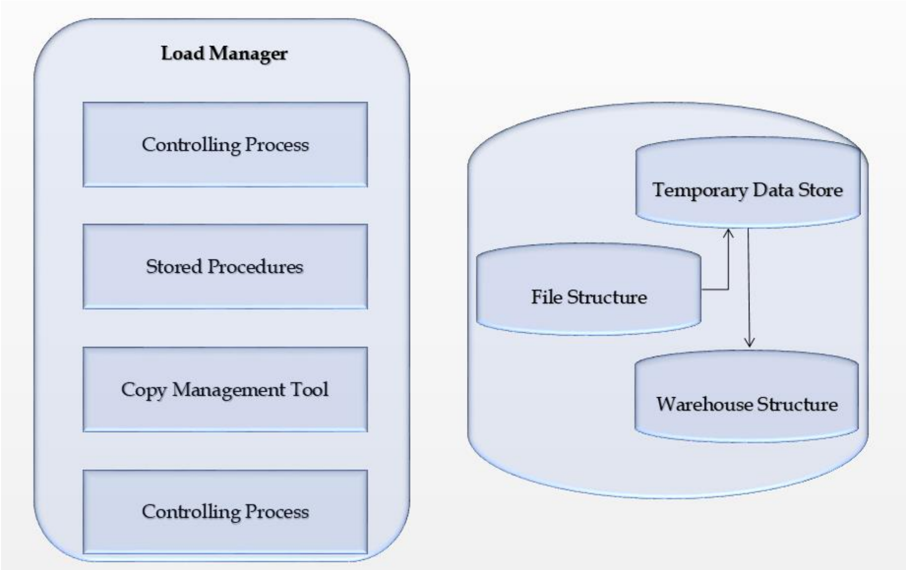

Figure 5: Load Manager Architecture

Functions of Load Manager

- The load manager does perform the following functions: -
- **Extract data from the source system**:The data is extracted from the operational databases or the external information providers.
- Gateways are the application programs that are used to extract data.
- **Fast load the extracted data into a temporary data store:**It is more effective to load the data into a relational database before applying transformations and checks.
- Perform simple transformations into a structure similar to the one in the data warehouse.
- Suppose we are loading the EPOS sales transaction,we need to perform the following checks:

➢ Strip out all the columns that are not required within the warehouse.

➢ Convert all the values to required data types.

Warehouse Manager

The warehouse manager is responsible for the warehouse management process. It consists of the third-party system software, C programs, and shell scripts. The size and complexity of a warehouse manager vary between specific solutions.

Components

A warehouse manager includes the following: −

- The controlling process
- Stored procedures or C with SQL
- Backup/Recovery tool
- SQL scripts

The architecture of Warehouse Manager

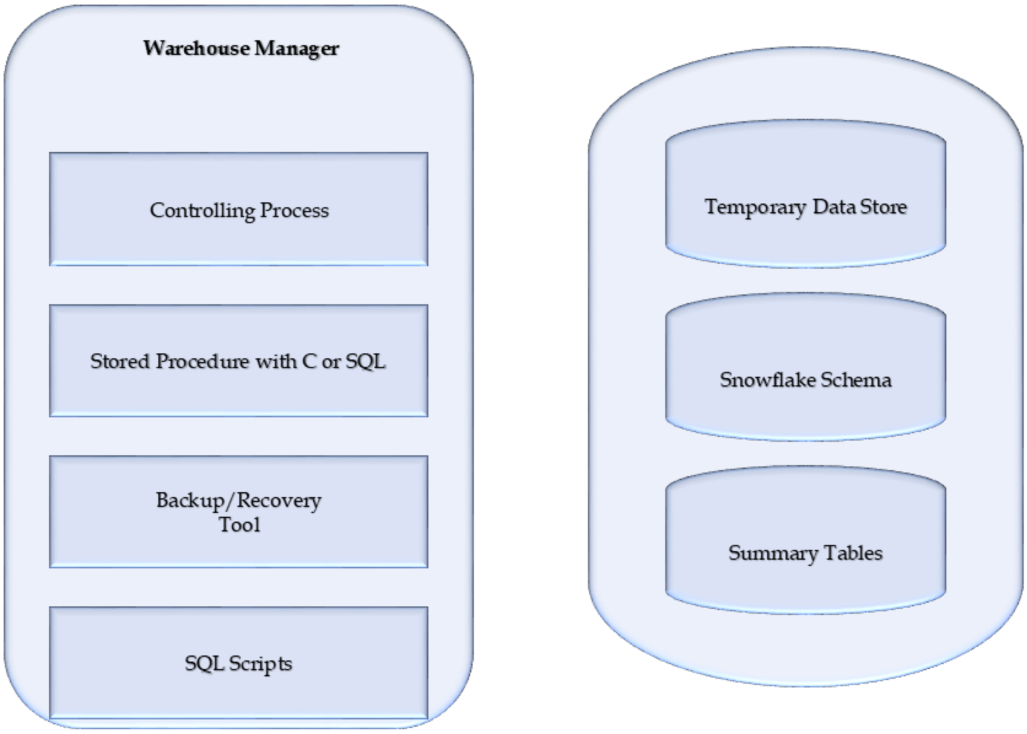

Figure 6: Warehouse Manager Architecture

Functions of Warehouse Manager

- Analyzes the data to perform consistency and referential integrity checks.
- Creates indexes, business views, partition views against the base data.
- Generates new aggregations and updates the existing aggregations.
- Generates normalizations.
- Transforms and merges the source data of the temporary store into the published data warehouse.
- Backs up the data in the data warehouse.
- Archives the data that has reached the end of its captured life.

3.3 Query Manager

The query manager is responsible for directing the queries to suitable tables. By directing the queries to appropriate tables, it speeds up the query request and response process. Besides, the query manager is responsible for scheduling the execution of the queries posted by the user. A query manager includes the following components :

- Query redirection via C tool or RDBMS
- Stored procedures
- Query management tool
- Query scheduling via C tool or RDBMS
- Query scheduling via third-party software.

Functions of Query Manager

- It presents the data to the user in a form they understand.

- It schedules the execution of the queries posted by the end-user.

- It stores query profiles to allow the warehouse manager to determine which indexes and aggregations are appropriate.
- Performs operations associated with management of user queries
- Constructed using vendor end-user data access tools, data warehouse monitoring tools, database facilities, and custom-built programs
- Complexity determined by facilities provided by end-user access tools and database
- Operations:
  - Directing queries to appropriate tables
  - Scheduling execution of queries

3.4 End-user Access Tools

There are five main groups of access tools:–

- Data reporting and query tools
- Application development tools
- Executive information system (EIS) tools
- Online analytical processing (OLAP) tools
- Data mining tools

Data Reporting and Query Tools

- The principal purpose of Data Warehousing is to provide information to business users for strategic decision-making. Different types of users engage in different types of decision

  support activities and therefore require different types of tools.When it comes time to start creating reports out of the data in your warehouse and to start making decisions with this data you are going to need to have a good query tool.Managed query tools shield end users from the Complexities of SQL and database structure by inserting a meta-layer between the user and the database.

Features for Query Tools

- **Cross-Browsing of Dimension Attributes**:A real dimension table, such as a list of all of your products or customers, takes the form of a large dimension table with many, many attributes (fields). Cross-browsing, on the other hand, refers to the capability of a query tool to present the valid values of the product brand, subject to a constraint elsewhere on that dimension table.
- **Open Aggregate Navigation:**Aggregate navigation is the ability to automatically choose pre-stored summaries, or aggregates, in the course of processing a user's SQL requests. Aggregate navigation must be performed silently and anonymously
- **Multipass SQL:**Breaking a single complex request into several small requests is called multipass SQL. It also allows drilling across several conformed data marts in different databases, in which the processing of a single galactic SQL statement would otherwise be impossible.
- **Semi-Additive Summations:**Semi Additive measures are values that you can summarize across any related dimension except time.Stock levels however are semi-additive; if you had 100 in stock yesterday, and 50 in stock today, you're total stock is 50, not 150. It doesn't make sense to add up the measures over time, you need to find the most recent value.
- **Show Me What Is Important:**Your query tools must help you automatically sift through the data to show you only what is important. At the low end, you simply need to show data rows in your reports that meet certain threshold criteria.

- **Behavioral Studies:**An interesting class of applications involves taking the results of a previous report or set of reports and then using these results over and over again at a later time.

Executive Information System (EIS) Tools

An Executive information system, also known as an Executive support system, is a type of management support system that facilitates and supports senior executive information and decision-making needs. It provides easy access to internal and external information relevant to organizational goals.

Reporting Tools

Reporting Tools can be divided into two categories:

Production Reporting Tools: These tools let companies generate regular operational reports or support high-volume batch jobs, such as calculating and printing paychecks. Production Reporting Tools include 3GLs such as COBOL, specialized 4GL, such as Information Builders, Inc’s Focus, and high-end client/ server tools such as MITI's SQR.

Desktop Report Writers: Report writers are inexpensive desktop tools designed for end-users. A product such as Crystal Reports, lets users design and run reports without having to rely on the IS Department.

OLAP Tools

Online Analytical Processing Server (OLAP) is based on the multidimensional data model. It allows managers, and analysts to get an insight into the information through fast, consistent, and interactive access to information.

Classification of OLAP Tools

MOLAP: It stands for Multidimensional Online Analytical Processing. It stores data in multidimensional arrays and requires pre-computation and storage of information in the cube.Some of the tools for that are:

- **IBM Cognos:** It provides tools for reporting, analysis, monitoring of events and metrics.
- **SAP NetWeaver BW:** It is known as SAP NetWeaver Business Warehouse. Just like IBM Cognos, It also delivers reporting, analysis, and interpretation of business data. It runs on Industry-standard RDBMS and SAP’s HANA in-memory DBMS.
- **Microsoft Analysis Services:** Microsoft Analysis Services is used by Organisations to make sense of data that is spread out in multiple databases or it is spread out in a discrete form.
- **MicroStrategy Intelligence Server**: MicroStrategy Intelligence Server helps the business to standardize themselves on a single open platform which in return will reduce their maintenance and operating cost.

MOLAP products are the commercial Hyperion Ebasse (www.hyperion.com) and the Applix TM1 (www.applix.com), as well as Palo (www.opensourceolap.org), which is an open-source product.

ROLAP: The ‘R’ in ROLAP stands for Relational. So, the full form of ROLAP becomes Relational Online Analytical Processing. The salient feature of ROLAP is that the data is stored in relational databases. Some of the top ROLAP is as follows:

- IBM Cognos
- SAP NetWeaver BW
- Microsoft Analysis Services
- Essbase
- Jedox OLAP Server

- SAS OLAP Server

ROLAP engines include the commercial IBM Informix Metacube (www.ibm.com) and the Micro-strategy DSS server (www.microstrategy.com), as well as the open-source product Mondrian (mondrian.sourceforge.net).

HOLAP: It stands for Hybrid Online Analytical Processing. So, HOLAP bridges the shortcomings of both MOLAP and ROLAP by combining their capabilities. Now how does it combine? It combines data by dividing data of the database between relational and specialized storage. Some of the top HOLAP is as follows:

Example IBM Cognos

SAP NetWeaver BW

Data Mining Tools

Data Mining tools have the objective of discovering patterns/trends/groupings among large sets of data and transforming data into more refined information. They Provide insights into corporate data that are not easily discerned with a managed query or OLAP tools. Tools usea variety of statistical and AI algorithm to analyze the correlation of variables in data. It investigates interesting patterns and their relationship

Figure 7: Data Mining Tools

Application Development Tools

Sometimes built-in graphical and analytical tools do not satisfy the analytical needs of an organization. In such cases, custom reports are developed using Application development tools.

3.5 Types of data in Data Warehouse

The data within the specific warehouse itself has a particular architecture with the emphasis on various levels of summarization, as shown in the figure:

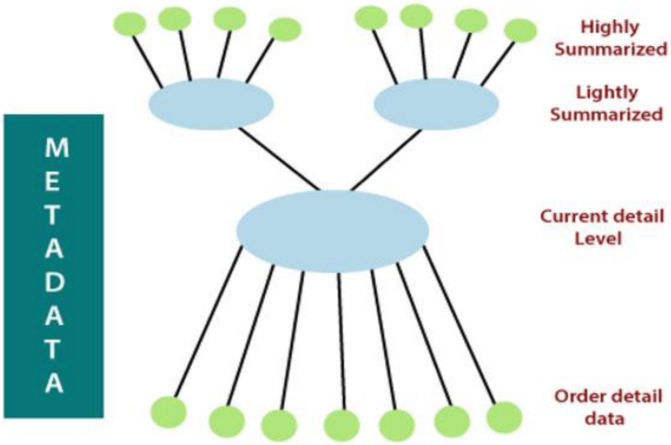

Figure 8:Structure of data inside Data Warehouse

Four types of data arestored in a data warehouse:

- older detail data,
- current detail data,
- lightly summarized data
- highly summarized data.

Older Detail Data:Older detail data represents data that is not very recent, maybe as old as ten years or longer.It is voluminous and most frequently stored on mass storage such as tape, although more expensive disk storage may be used. Its level of detail is consistent with current detail data

but due to its long time horizon is typically migrated to a "less-expensive" alternate storage medium.

Current Detail Data: This data represents data of a recent nature and always has a shorter time horizon than older detail data. Although it can be voluminous, it is almost always stored on a disk to permit faster access. The current detail record is central in importance as it:

- Reflects the most current happenings, which are commonly the most stimulating.
- It is numerous as it is saved at the lowest method of the Granularity.
- It is always (almost) saved on disk storage, which is fast to access but expensive and difficult to manage.

Lightly summarized data: Lightly summarized data represents data distilled from current detail data. It is summarized according to some unit of time and always resides on disk. This data extract from the low level of detail found at the current, detailed level and usually is stored on disk storage. When building the data warehouse have to remember what unit of time is the summarization done over and also the components or what attributes the summarized data will contain.

Highly summarized data is compact and directly available and can even be found outside the warehouse. Highly summarized data represents data distilled from lightly summarized data. It is always compact and easily accessible and resides on a disk.A final component of the data warehouse is that of metadata. Metadata is best described as data about data. It provides information about the structure of a data warehouse as well as the various algorithms used in data summarizations. It provides a descriptive view, or "blueprint", of how data is mapped from the operational level to the data warehouse.

3.6 Data Archiving

Data warehouse archiving is the process of moving data that is not likely needed to conduct business operations from the data warehouse to a medium where it can be stored for long-term retention and retrieval if needed.Archive data consists of older data that remains important to the organization or must be retained for future reference.Data archives are indexed and have search capabilities, so files can be located and retrieved.Some archive systems treat archive data as read-only to protect it from modification, while other data archiving products enable writes, as well as reads.

WORM (write once, read many) technologies use media that is not rewritable.

Backup data

A data warehouse is a complex system and it contains a huge volume of data. Therefore it is important to back up all the data so that it becomes available for recovery in the future as per requirement.

Backup Terminologies

- **Complete backup** − It backs up the entire database at the same time. This backup includes all the database files, control files, and journal files.
- **Partial backup** − As the name suggests, it does not create a complete backup of the database. Partial backup is very useful in large databases because they allow a strategy whereby various parts of the database are backed up in a round-robin fashion on a day-to-day basis so that the whole database is backed up effectively once a week.
- **Cold backup** − Cold backup is taken while the database is completely shut down. In a multi-instance environment, all the instances should be shut down.
- **Hot backup** − Hot backup is taken when the database engine is up and running. The requirements of hot backup vary from RDBMS to RDBMS.
- **Online backup** − It is quite similar to hot backup.

	Backup	Archiving
What is it?	Protection for mission critical systems and live data	Searchable records of inactive data in a "steady state"
Why use it?	Recovery – backup restores systems after data loss, interruption, or disaster	Searching – allows interrogation of data for regulatory inspections and data investigations
What does it contain?	Several snapshots of the live system(s) captured on a time basis	One single repository of historical data indexed and quickly searchable

3.7 Metadata

- Metadata is data about data that defines the data warehouse. It is used for building, maintaining, and managing the data warehouse.In the Data Warehouse Architecture, meta-data plays an important role as it specifies the source, usage, values, and features of data warehouse data. It also defines how data can be changed and processed. For example, a line in the sales database may contain:

4030 KJ732 299.90

This is meaningless data until we consult the Meta that tell us it was

- Model number: 4030

- Sales Agent ID: KJ732
- Total sales amount of $299.90
- Therefore, Meta Data are essential ingredients in the transformation of data into knowledge.

Metadata is the final element of the data warehouses and is really of various dimensions in which it is not the same as file drawn from the operational data, but it is used as:-

- A directory to help the DSS investigator locate the items of the data warehouse.
- A guide to the mapping of record as the data is changed from the operational data to the data warehouse environment.
- A guide to the method used for summarization between the current, accurate data and the lightly summarized information and the highly summarized data, etc.

Metadata helps to answer the following questions:

- What tables, attributes, and keys does the Data Warehouse contain?
- Where did the data come from?
- How many times do data get reloaded?
- What transformations were applied with cleansing?

Categories of metadata

- **Technical MetaData**: This kind of Metadata contains information about the warehouse which is used by Data warehouse designers and administrators.
- **Business MetaData:** This kind of Metadata contains detail that gives end-users a way easy to understand the information stored in the data warehouse.

Data warehouse metadata include table and column names, their detailed descriptions, their connection to business meaningful names, the most recent data load date, the business meaning of a data item and the number of users that are logged in currently

3.8 Architecture Model

Architecture model is a method of defining the overall architecture of data communication, processing, and presentation that exist for end-clients computing within the enterprise. Applications gather detailed data from day-to-day operations. A data warehouse architecture is a method of defining the overall architecture of data communication processing and presentation that exists for end-clients computing within the enterprise. Each data warehouse is different, but all are characterized by standard vital components.

Types of Data Warehouse Architectures

Two-Tier Architecture

The requirement for separation plays an essential role in defining the two-tier architecture for a data warehouse system, as shown in fig:

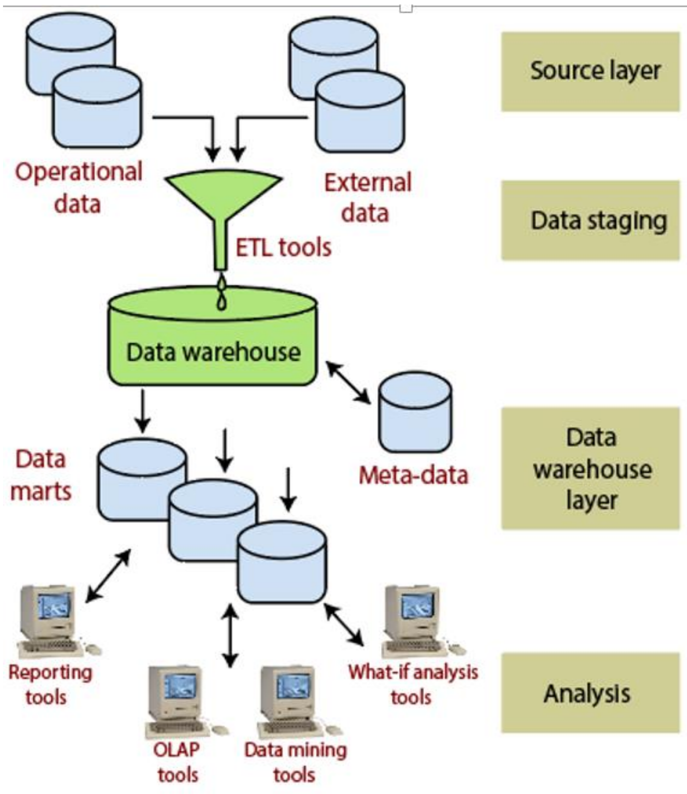

Figure 9:Two -Tier Data Warehouse Architecture

Although it is typically called two-layer architecture to highlight a separation between physically available sources and data warehouses, in fact, consists of four subsequent data flow stages:

Source layer: A data warehouse system uses a heterogeneous source of data. That data is stored initially to corporate relational databases or legacy databases, or it may come from an information system outside the corporate walls.

Data Staging: The data stored to the source should be extracted, cleansed to remove inconsistencies and fill gaps, and integrated to merge heterogeneous sources into one standard schema. The so-named **Extraction**, **Transformation**, and **Loading Tools (ETL)** can combine heterogeneous schemata, extract, transform, cleanse, validate, filter, and load source data into a data warehouse.

Data Warehouse layer: Information is saved to one logically centralized individual repository: a data warehouse. The data warehouses can be directly accessed, but they can also be used as a source for creating data marts, which partially replicate data warehouse contents and are designed for specific enterprise departments. Meta-data repositories store information on sources, access procedures, data staging, users, data mart schema, and so on.

Analysis: In this layer, integrated data is efficiently, and flexible accessed to issue reports, dynamically analyze information, and simulate hypothetical business scenarios. It should feature aggregate information navigators, complex query optimizers, and customer-friendly GUIs.

Three-Tier Architecture

The three-tier architecture consists of the source layer (containing multiple source systems), the reconciled layer, and the data warehouse layer (containing both data warehouses and data marts). The reconciled layer sits between the source data and data warehouse.

The main advantage of the **reconciled layer** is that it creates a standard reference data model for a whole enterprise. At the same time, it separates the problems of source data extraction and integration from those of the data warehouse population. In some cases, the **reconciled layer** is also directly used to accomplish better some operational tasks, such as producing daily reports that cannot be satisfactorily prepared using the corporate applications or generating data flows to feed external processes periodically to benefit from cleaning and integration.

This architecture is especially useful for extensive, enterprise-wide systems. A disadvantage of this structure is the extra file storage space used through the extra redundant reconciled layer. It also makes the analytical tools a little further away from being real-time.

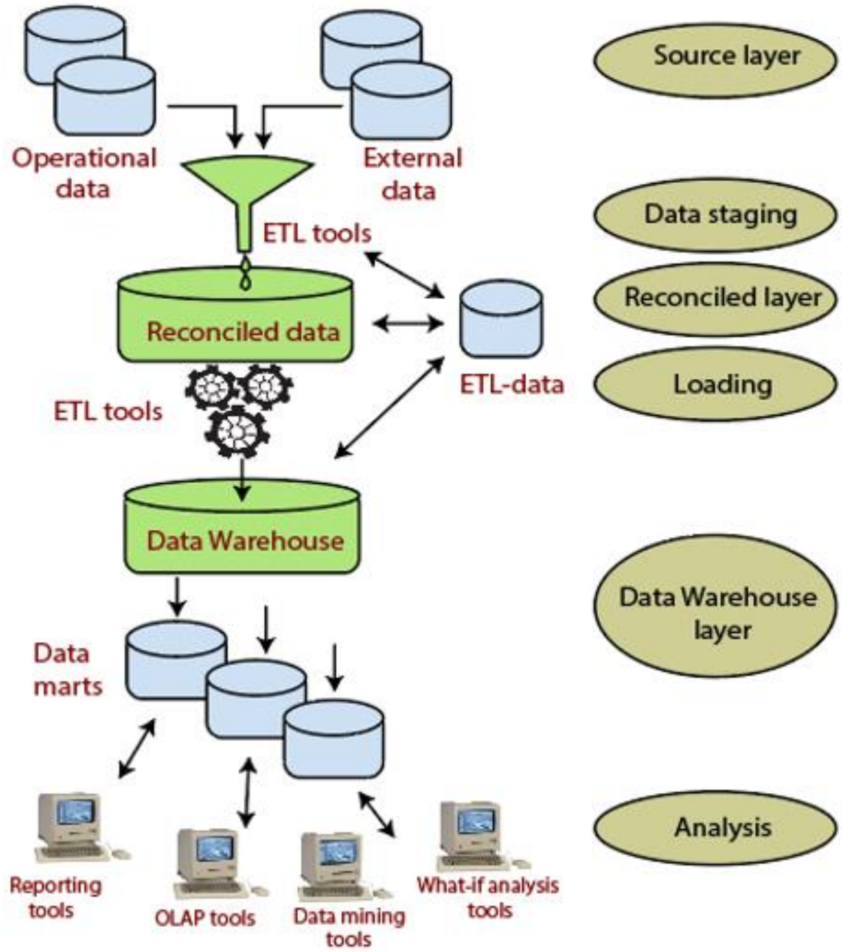

Figure 10: Three Tier Data Warehouse Architecture

4-Tier architecture

User: At the end-**user layer**, **data** in the ODS, **data warehouse**, and **data** marts can be accessed by using a variety of tools such as query and reporting tools, **data** visualization tools, and analytical applications.

Presentation layer: Its functions contain receiving data inputted, interpreting users' instructions, and sending requests to the data services layer, and displaying the data obtained from the data services layer to users by the way they can understand. It closest to users and provide an interactive operation interface.

Discuss various factors play vital role to design a good data warehouse.

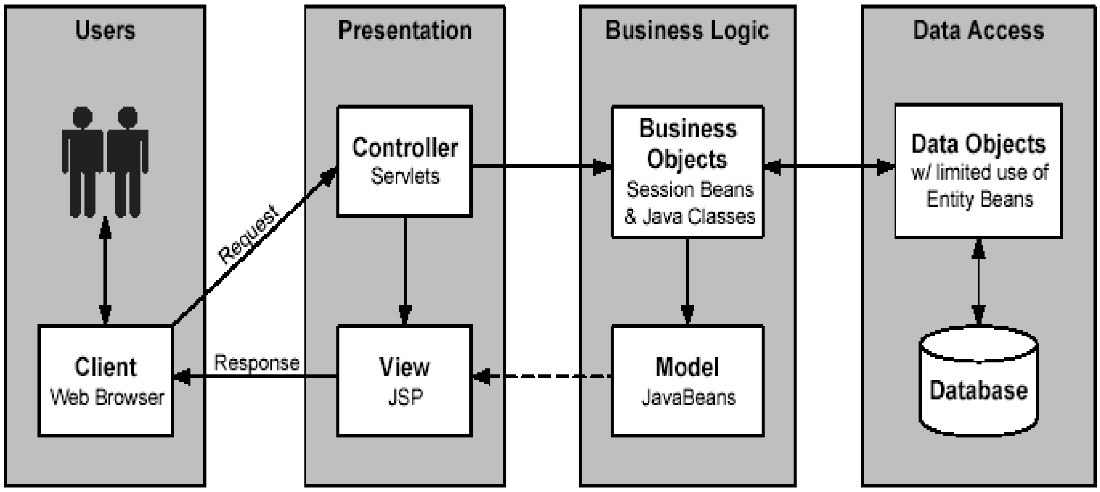

Figure 11:Four Tier Data Warehouse Architecture

Business logic:It is located between the PL and data access layer playing a connecting role in the data exchange.The layer's concerns are focused primarily on the development of business rules, business processes, and business needs related to the system.It's also known as the domain layer.

Data Access: It is located in the innermost layer that implements persistence logic and responsible for access to the database.Operations on the data contain finding, adding, deleting, modifying, etc. This level works independently, without relying on other layers.DAL extracts the appropriate data from the database and passes the data to the upper.

Summary

- OLAP servers may adopt a relational OLAP (ROLAP), a multidimensional OLAP (MOLAP), or a hybrid OLAP (HOLAP) implementation.
- Data warehousing is the consolidation of data from disparate data sources into a single target database to be utilized for analysis and reporting purposes.
- The primary goal of data warehousing is to analyze the data for business intelligence ⌘⌘purposes. For example, an insurance company might create a data warehouse to capture policy data for catastrophe exposure.
- The data is sourced from front-end systems that capture the policy information into the ⌘⌘data warehouse.
- The data might then be analyzed to identify windstorm exposures in coastal areas prone to ⌘⌘hurricanes and determine whether the insurance company is overexposed.
- The goal is to utilize the existing information to make accurate business decisions.

Keywords

Data Sources: Data sources refer to any electronic repository of information that contains data of interest for management use or analytics.

Data Warehouse Architecture: It is a description of the elements and services of the warehouse, with details showing how the components will fit together and how the system will grow over time.

Data Warehouse: It is a relational database that is designed for query and analysis rather than for transaction processing.

Job Control: This includes job definition, job scheduling (time and event), monitoring, logging, exception handling, error handling, and notification.

Metadata: Metadata, or "data about data", is used not only to inform operators and users of the data warehouse about its status and the information held within the data warehouse

Self Assesment

1) OLTP stands for

(a) On Line Transactional Processing

(b) On Link Transactional Processing

(c) On Line Transnations Processing

(d) On Line Transactional Program

2) ROLAP stands for

(a) Relational On Line Transition Processing

(b) Relative On Line Transactional Processing

(c) Relational On Line Transactional Processing

(d) Relational On Line Transactional Program

3) The data from the operational environment enter …………………… of data warehouse.

A) Current detail data

B) Older detail data

C) Lightly Summarized data

D) Highly summarized data

4) ………………………are designed to overcome any limitations placed on the warehouse by the nature of the relational data model.

A) Operational database

B) Relational database

C) Multidimensional database

D) Data repository

5) Data warehouse contains ________data that is seldom found in the operational environment Select one:

a)informational

b)normalized

c)denormalized

d)summary

6) _______ are numeric measurements or values that represent a specific business aspect or activity Select one:

a)Dimensions

b)Schemas

c)Facts

d)Tables

7) _____describes the data contained in the data warehouse Select one:

a)Relational data

b)Operational data

c)Informational data

d)Meta data

8) Dimensionality refers to Select one:

a)Cardinality of key values in a star schema

b)The data that describes the transactions in the fact table

c)The level of detail of data that is held in the fact table

d)The level of detail of data that is held in the dimension table

9) Business Intelligence and data warehousing is used for ………….

A) Forecasting

B) Data Mining

C) Analysis of large volumes of product sales data

D) All of the above

10) Data warehouse is a kind of analytical tool used in ……………..

11) An operational system is which of the following?

a)A system that is used to run the business in real time and is based on historical data.

b)A system that is used to run the business in real time and is based on current data.

c)A system that is used to support decision making and is based on current data.

d)A system that is used to support decision making and is based on historical data.

12) Decision support systems (DSS) is

a)A family of relational database management systems marketed by IBM

b)Interactive systems that enable decision makers to use databases and models on a computer in order to solve ill-structured problems

c)It consists of nodes and branches starting from a single root node. Each node represents a test, or decision

d)None of these

13) …………….. stores the data based on the already familiar relational DBMS technology.

14) Which of the following features usually applies to data in a data warehouse?

A.Data are often deleted

B.Most applications consist of transactions

C.Data are rarely deleted

D.Relatively few records are processed by applications

15) The following is true of three-tier data warehouses:

a)Once created, the data marts will keep on being updated from the data warehouse at periodic times

b)Once created, the data marts will directly receive their new data from the operational databases

c)The data marts are different groups of tables in the data warehouse

d)A data mart becomes a data warehouse when it reaches a critical size

Review Questions

1. What is a data warehouse? How is it better than traditional information-gathering techniques?
2. Describe the data warehouse environment.
3. List and explain the different layers in the data warehouse architecture.
4. Differentiate between ROLAP and MOLAP.
5.Describe three-tier data warehouse architecture in detail.

Answers: Self Assesement

1.	A	2.	C	3.	A	4.	C	5.	D
6.	C	7.	D	8.	B	9.	D	10.	Bussiness Area
11.	B	12.	B	13.	ROLAP	14.	C	15.	A

Further Readings

A. K. Jain and R. C. Dubes, *Algorithms for Clustering Data*, Prentice Hall, 1988.

Alex Berson, *Data Warehousing Data Mining and OLAP*, Tata McGraw Hill, 1997

Alex Berson, Stephen J. Smith, *Data warehousing, Data Mining & OLAP*, Tata McGraw Hill, Publications, 2004.

Alex Freitas and Simon Lavington, *Mining Very Large Databases with Parallel Processing*, Kluwer Academic Publishers, 1998.

J. Ross Quinlan, *C4.5: Programs for Machine Learning*, Morgan Kaufmann Publishers, 1993.

Jiawei Han, MichelineKamber, *Data Mining – Concepts and Techniques*, Morgan Kaufmann Publishers, First Edition, 2003.

Matthias Jarke, Maurizio Lenzerini, YannisVassiliou, PanosVassiliadis, *Fundamentals of Data Warehouses*, Publisher: Springer

Michael Berry and Gordon Linoff, *Data Mining Techniques (For Marketing, Sales, and Customer Support)*, John Wiley & Sons, 1997.

Michael J. A. Berry, Gordon S Linoff, *Data Mining Techniques*, Wiley Publishing Inc, Second Edition, 2004.

www.en.wikipedia.org

www.web-source.net

www.webopedia.com